

ASTREL: Prototyping Shape-changing Interface with Variable Stiffness Soft Robotics Module

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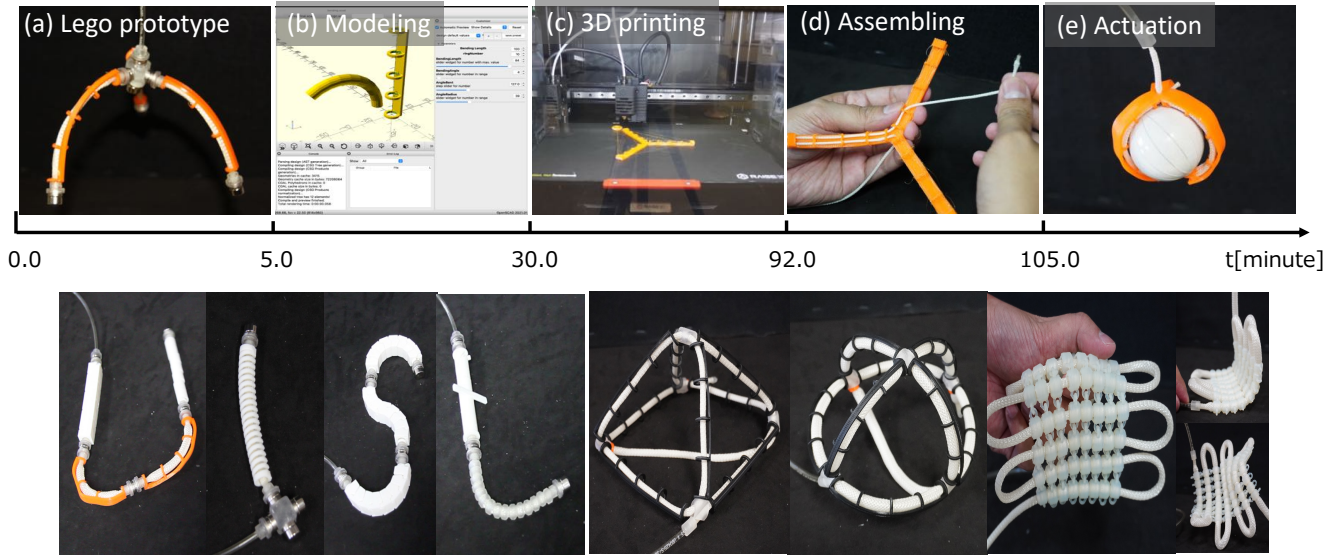


Figure 1: Upper image: Fabrication workflow, lower image: Prototype example

ABSTRACT

Prototyping a shape-changing interface is challenging because it requires knowledge of both electronics and mechanical engineering. In this study, we introduced a prototyping platform using a soft pneumatic artificial muscle (PAMs) and modular 3D printed reinforcement. To facilitate a wide variety of applications we propose six types of reinforcement modules capable of either shape deformation and/or variable stiffness. Users can create an approximate prototype using lego-built modules with magnetic connectors. A modeling toolkit can then be used to recreate and customize the prototype structure. After 3D printing, the shape-changing interface can be assembled by threading the PAMs through holes in the reinforcement. We envision that this prototyping platform can be useful in shape-changing interface exploration, where researchers can create working prototypes easily, rapidly, and at a low cost.

CCS CONCEPTS

• **Human-centered computing** → **Displays and images**; *Mixed / Interaction Design*.

KEYWORDS

Shape-changing Interface, Soft Robotics, Variable Stiffness, Modular Interface, Fabrication toolkit

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1 INTRODUCTION

Three-dimensional (3D) printing has become a key technology in the fabrication of shape-changing interfaces. Several studies have demonstrated the capability of directly printing pneumatic and hydraulic actuated shape-changing interfaces [2] [9]. Although both techniques provide convenient tools for fabrication, they require either a high-end printer or a modified 3D printer. Another limitation is that the durability and reliability of actuators are highly dependent on the 3D printer and materials used. In prior works, we introduced a prototyping technique that utilized an unmodified

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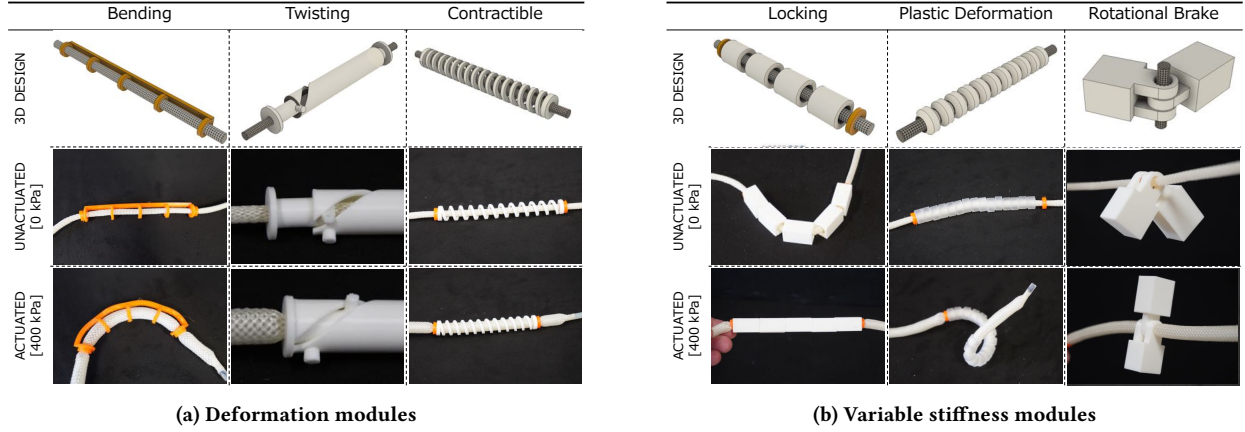


Figure 2: Six type of modules

desktop 3D printer and common materials, such as PLA and TPU[5]. We also used a premanufactured PMAs actuator with high durability and efficiency (40N contraction force). In addition, 3D printed substrate are only used for reinforcement structures; hence, less material and printing time are required. However, our technique has a shortcoming in the requirements of the assembly of PAMs and 3D printed structures.

Pneumatic artificial muscles(PAMs) such as McKibben actuators, since patented in the 1950s have been advanced into small, lightweight, and flexible actuators. Recent studies in soft McKibben actuators have proposed their application in soft robotics and shape-changing interfaces[8] [1]. Although both of these works also utilized 3D printed reinforcement in their mechanism. Our focus is on the variety of 3D printed reinforcement designs which include shape deformation and variable stiffness properties. Our fabrication technique also has an advantage where both the PAMs and reinforcement modules are reusable and replaceable.

Variable stiffness is a beneficial feature for shape-changing interface because it can exponentially increase the interaction possibility. The common method of variable stiffness is the vacuum jamming technique, which includes granular, layer, and fiber jamming [7], [4]. In this paper, we proposed a convenient control system, in which both shape deformation and variable stiffness are powered by compressed air pressure.

Modularity in shape-changing interface offers great advantages, such as reducing fabrication and maintenance costs, where independent modules can be easily replaced. It also offers reconfigurability, which allows the user interface to be scaled or rearranged to adapt to the environment. Previous studies have proposed modular shape-changing interfaces and soft robots with magnetic connection [6], [3]. We took inspiration from previous studies and combined it with our proposed variable stiffness reinforcement to further increase the shape and interaction possibilities.

2 DESIGN AND METHOD

We propose six types of modules that can be classified into deformation modules and variable stiffness modules. Fig. 2 shows the 3D model and difference between the actuated and unactuated states.

2.1 Deformation modules

Deformation is achieved by altering the contraction force using 3D printed reinforcements. There are three types of deformations:

- (1) **Bending module** is based on the bilayer actuation phenomenon. We 3D printed the bending reinforcement using flexible material(TPU95).
- (2) **Twisting module** is based on the screw mechanism, converting PAMs linear motion into rotation.
- (3) **Contractible module** is based on the spring structure. Although PAMs can contract without spring reinforcement, they add rigidity to the structure and store compression energy to exert opposing forces.

2.2 Variable stiffness modules

Variable stiffness is achieved by altering both the contraction and expansion forces of the PAMS. The three types of variable stiffness are:

- (1) **Locking module** is based on the beaded jamming phenomenon. We use the contraction force of the PAMs to tension an array of hollow modules and interlock them with each other.
- (2) **Plastic deformation module** has unique behavior which allows the structure to retain its shape after being bent by an external force.
- (3) **Rotational brake module** is based on a hinge structure with PAMs as shafts. When PAMs are unactuated, the hinge can rotate freely. However, when the PAMs are expanded inside the hinge, they rub together and resist rotation.

2.3 Fabrication workflow

Lego prototype : We adopted the LEGO-built assembling method, where users can conveniently add and replace various modules. To connect each module, we attached a ring-shaped magnetic connector capable of withstanding high-pressure air(0.4MPa) for PAMs actuation. To facilitate the branching of the air channel, we created a cube and prism-shaped connector with up to four branches. To plug the unused air channels, we used a cylindrical magnet. In

Table 1: Force and stiffness characteristic

	Max. Force (N)	Max. Stiffness (N/mm)
Bending	4.1	
Twisting	4.3	
Contractible	21.4	
Locking		0.85
Plastic Deformation		0.86
Rotational brake		0.95

addition to the six types of modules, we created some variations, such as locking with an arched shape and print in-place hinges to limit the deformation.

Modeling :Based on this prototype, the user can create an STL model for 3D printing. We provide a simple toolkit where users can easily adjust the length and number of reinforcement divisions. Users can also increase the number of PAMs threaded onto the reinforcement to scale the force.

3D printing and assembly : For final fabrication, users can use both FDM and SLA 3D printers. The material used was mostly rigid filaments such as PLA and ABS; however, for bending modules, we used the flexible filament TPU95. After 3D printing, the user assembles the reinforcement similar to bead craftworks. Here manual work of threading PAMs through holes on reinforcements is required.

Actuation control : Both the deformation state and stiffness level is controlled by air pressure. The air pressure controlled by the electro-pneumatic regulator ranged from 0.4 MPa. Table 1 shows the Max. deformation force and Max. stiffness of each module.

3 FEATURES

3.1 Module Combination

Our prototyping platform provided six reinforcement module with elongated shapes along the PAMs. Owing to the simple one-dimensional shape, each module can be arranged in series and parallel. Moreover, modularity grants flexibility to the mixing arrangements between the features.

3.2 Scalability

The conventional method of increasing contraction force of PAMs is to use larger PAMs with a larger cross-sectional area. We introduce a scaling technique that uses only one long PAMs filament arranged in a zigzag loop. This technique allows convenient implementation, with less effort of creating multiple air-supply connections.

3.3 Fabric-like structure

Fig. 1(bottom right) shows an example implementation of structured fabric with variable stiffness. The fabric is soft and flexible in the normal state, and malleable when actuated. It consists of a plastic deformation module that is linked with chain structure in parallel arrangement. It can be bent both head-on and sideways, it also can be twisted and will retain the shapes when actuated.

4 DISCUSSION AND FUTURE WORK

Currently, we only develop a simple toolkit to modify one module. In the future, an integrated software that can simulate deformation and recreate shape from existed models will be beneficial. The current toolkit is also only capable of customizing the dimensions. In the future, toolkits could also adjust the partial stiffness and shape deformation by varying the reinforcement thickness, hole size, and number of restraining rings.

In this study, we did not implement sensors for application. However, for interaction purposes, currently installed air-pressure sensors and additional sensors such as acoustic, optical, and contact sensors can be utilized. The sensor can be inserted into the 3D printed substrate or embedded into the PAMs filament. It would also be interesting to detect the shape through magnetic sensor.

We believe that in the current stage of shape-changing interface development, exploring applications and interaction will motivate progress in core technology such as materials, actuation, sensing, and energy.

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